DIAGNOSTIC REASONING ASSISTANT

HYBRID ML + RETRIEVAL (RAG) FOR DIFFERENTIAL DIAGNOSIS

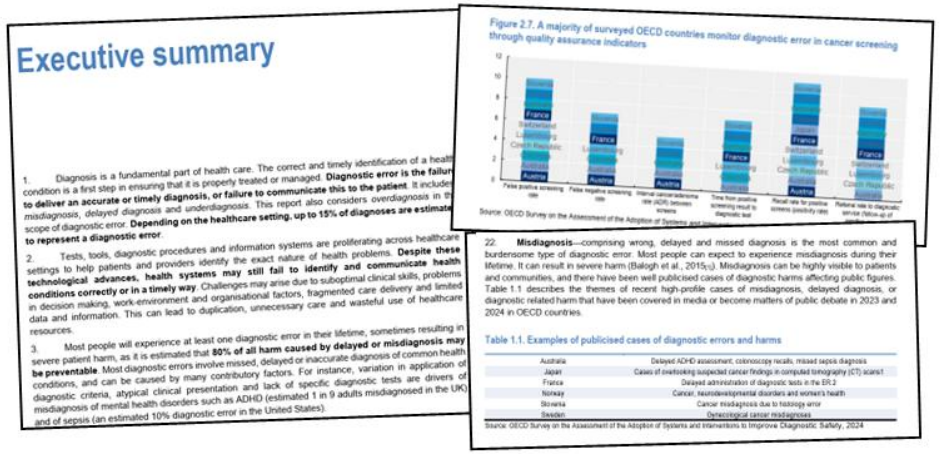
Dataset: DDXPlus (1.3M synthetic cases)

Julia Parnis ●Ironhack Bootcamp ●2026

Why this matters

Diagnostic errors are common — and AI can help, if designed safely

- OECD (Europe)



Executive summary

1. Diagnosis is a fundamental part of health care. The correct and timely identification of a health condition is a first step in ensuring that it is properly treated or managed. **Diagnostic error is the failure to deliver an accurate or timely diagnosis, or failure to communicate this to the patient.** It includes misdiagnosis, delayed diagnosis and underdiagnosis. This report also considers overdiagnosis in the scope of diagnostic error. **Depending on the healthcare setting, up to 15% of diagnoses are estimated to represent a diagnostic error.**

2. Tests, tools, diagnostic procedures and information systems are proliferating across healthcare settings to help patients and providers identify the exact nature of health problems. **Despite these technological advances, health systems may still fail to identify and communicate health conditions correctly or in a timely way.** Challenges may arise due to suboptimal clinical skills, problems in decision making, work-environment and organisational factors, fragmented care delivery and limited data and information. This can lead to duplication, unnecessary care and wasteful use of healthcare resources.

3. Most people will experience at least one diagnostic error in their lifetime, sometimes resulting in severe patient harm, as it is estimated that **80% of all harm caused by delayed or misdiagnosis may be preventable.** Most diagnostic errors involve missed, delayed or inaccurate diagnosis of common health conditions, and can be caused by many contributory factors. For instance, variation in application of diagnostic criteria, atypical clinical presentation and lack of specific diagnostic tests are drivers of misdiagnosis of mental health disorders such as ADHD (estimated 1 in 9 adults misdiagnosed in the UK) and of sepsis (an estimated 10% diagnostic error in the United States).

Figure 2.7. A majority of surveyed OECD countries monitor diagnostic error in cancer screening through quality assurance indicators

Source: OECD Survey on the Assessment of the Adoption of Systems and Interventions to Improve Diagnostic Safety, 2024

22. **Misdiagnosis**—comprising wrong, delayed and missed diagnosis is the most common and burdensome type of diagnostic error. Most people can expect to experience misdiagnosis during their lifetime. It can result in severe harm (Balogh et al., 2015[22]). Misdiagnosis can be highly visible to patients and communities, and there have been well publicised cases of diagnostic harms affecting public figures. Table 1.1 describes the themes of recent high-profile cases of misdiagnosis, delayed diagnosis, or diagnostic related harm that have been covered in media or become matters of public debate in 2023 and 2024 in OECD countries.

Table 1.1. Examples of publicised cases of diagnostic errors and harms

Australia	Delayed ADHD assessment; colonoscopy recalls; missed sepsis diagnosis
Japan	Cases of overlooking suspected cancer findings in computed tomography (CT) scans
France	Delayed administration of diagnostic tests in the ER
Norway	Cancer, neurodevelopmental disorders and women's health
Slovenia	Cancer misdiagnosis due to histology error
Sweden	Gynecological cancer misdiagnoses

Source: OECD Survey on the Assessment of the Adoption of Systems and Interventions to Improve Diagnostic Safety, 2024

- England Primary Care

ORIGINAL RESEARCH

OPEN ACCESS

Incidence, origins and avoidable harm of missed opportunities in diagnosis: longitudinal patient record review in 21 English general practices

Sudeh Cheraghi-Sohi,[1,2] Fiona Holland,[2,3] Hardeep Singh,[4] Avril Danczak,[5] Aneez Esmail,[2] Rebecca Lauren Morris,[1] Nicola Small,[2] Richard Williams,[1] Carl de Wet,[6] Stephen M Campbell,[1,2] David Reeves[2,3]

► Additional supplemental material is published online only. To view, please visit the journal online (http://dx.doi.org/10.1136/bmjqs-2020-012594).

For numbered affiliations see end of article.

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ABSTRACT

Background Diagnostic error is a global patient safety priority.

Objectives To estimate the incidence, origins and avoidable harm of diagnostic errors in English general practice. Diagnostic errors were defined as missed opportunities to make a correct or timely diagnosis based on the evidence available (missed diagnostic opportunities, MDOs).

Method Retrospective medical record reviews identified MDOs in 21 general practices. In each practice, two trained general practitioner reviewers independently conducted case note reviews on 100 randomly selected adult consultations performed during 2013–2014. Consultations where either reviewer identified an MDO were jointly reviewed.

caregivers, practitioners and health systems.[1] Preventable harm may occur when undiagnosed conditions remain untreated or when patients undergo unnecessary (or harmful) tests. Primary care underpins the UK health system, with an estimated 340 million patient consultations in England annually (or approximately 85% of all National Health Service doctor–patient encounters).[2] Diagnostic errors account for the greatest proportion of malpractice claims against family practitioners (general practitioners, GPs) in the UK (63%) incurring multiple costs.[3]

BMJ Qual Saf: first published as 10.1136/bmjqs-2020-012594 on 14 June 2021. Downloaded
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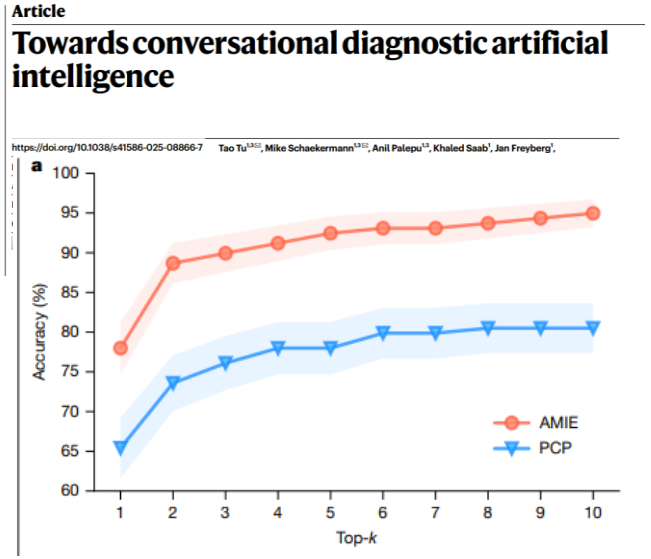
- AI Evidence



Article

Towards conversational diagnostic artificial intelligence

https://doi.org/10.1038/s41586-025-08866-7 Tao Tu[1,3], Mike Schaekermann[1,3], Anil Palepu[1,3], Khaled Saab[1], Jan Freyberg[1],

Up to 15%

of diagnoses may be inaccurate, delayed, or wrong (varies by setting).

~4.3%

confirmed missed diagnostic opportunities per 100 consultations (conservative).

↑ Top-k DDx

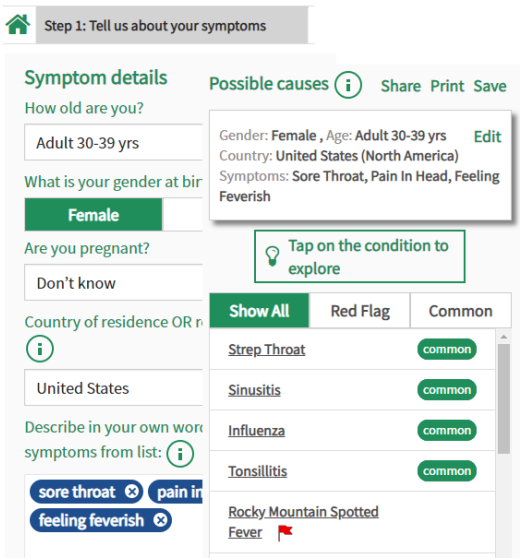
Conversational AI (AMIE) outperformed PCPs in OSCE-style simulated consultations.

Project focus: transparent top-k differential diagnosis + targeted follow-up questions (not a single "final answer").

Sources: OECD (2025) The Economics of Diagnostic Safety; Cheraghi-Sohi et al. BMJ Qual Saf (2021); Tu et al. Nature (2025).

Symptom checkers today

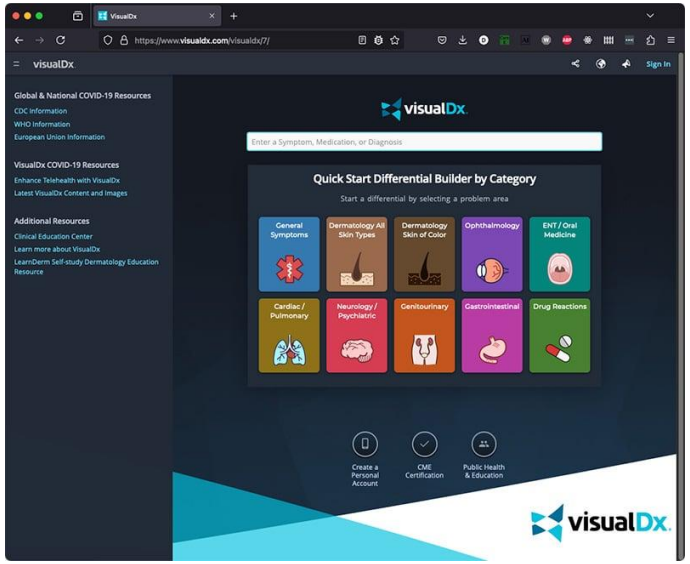
Useful, but often long lists + inconsistent accuracy



PATIENT-FACING

Isabel

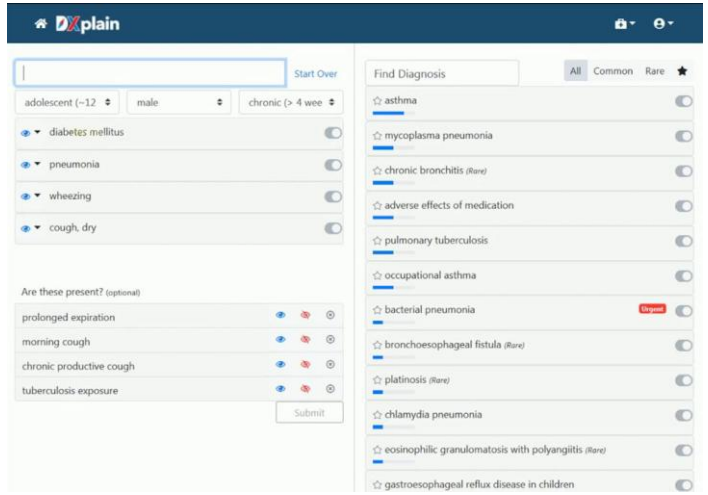
Broad symptom lists



CLINICIAN-FACING

VisualDx

Image-rich differential support



CLASSIC DDx

DXplain

Hypothesis generation

What research shows (BMJ audit of 23 symptom checkers):

- **34%:** Correct diagnosis ranked #1
- **58%:** Correct within top-20
- **57% :** Appropriate triage advice

Takeaway: these tools are often risk-averse → broad differentials.

We treat diagnosis as an iterative narrowing process:

rank → ask → update

(instead of listing 20+ conditions).

Source: Semigran et al., BMJ (2015) symptom checker audit (45 vignettes).

Our approach: combine ML ranking with evidence-grounded follow-up

🎯 GOAL

Build an educational diagnostic reasoning assistant that ranks differential diagnoses (Top-k), refines through iterative questioning and provides grounded explanations

Use ML for probability, use retrieval for explanation, and let each new answer refine the differential.

1 Initial evidence
Age, sex, initial evidence list

2 ML ranking
Top-k diseases + probabilities

3 RAG retrieval
Guidelines / literature + citations

4 Next question
Ask the most discriminating follow-up

5 Update
Positive / negative / unknown → rerank

What the ML layer contributes now

Efficient ranking over a large evidence space: it turns a raw symptom list into a probabilistic shortlist that can be updated every time new evidence arrives.

What the RAG layer adds later

Human-readable rationale, trustworthy citations, and better question wording. The aim is not to replace ranking, but to make the next step understandable and defensible.

Design principle: broad list in → smaller, better-justified list out

Why DDXPlus?

It matches an iterative differential diagnosis workflow

Dataset	What it provides	Gap for my goal
DDXPlus	Symptoms/antecedents + true pathology + differential diagnosis list (ranking target)	Synthetic
MIMIC-IV	Real ICU/ED Electronic Health Record (EHR) (codes, labs, notes)	No explicit differential list per encounter; messy missingness
Synthea	Synthetic EHR timelines (longitudinal)	Not designed for symptom-question DDx loop

Why it fits this project:

- Explicit differential diagnosis lists (ranking)
- Multi-type evidence schema (binary/categorical/multi-choice)
- Large scale for robust baselines

DDXPlus structure

Three linked files: patients ↔ evidences ↔ conditions

Patients (main table)

- AGE, SEX
- INITIAL_EVIDENCE
- EVIDENCES (list)
- PATHOLOGY (label)
- DIFFERENTIAL_DIAGNOSIS (list)

EXAMPLE ROW

```
AGE: 18
SEX: "M"
PATHOLOGY: "URTI"

INITIAL_EVIDENCE:
  ["E_54 @ V_161", "E_55 @ V_183"]

EVIDENCES:
  ["E_9", "E_27", "E_50",  ...]

DIFFERENTIAL_DIAGNOSIS:
  ["Bronchitis", 0.1917...],
  ["Pneumonia", 0.1757...]
```

Evidences (map)

- question text
- type: B / C / M
- default_value
- possible values
- is_antecedent

EXAMPLE EVIDENCE: E_91

```
{
  "name": "E_91",
  "code_question": "E_91",
  "question_fr":
    "Avez-vous objectivé...",
  "question_en":
    "Do you have a fever?",
  "is_antecedent": false,
  "default_value": 0,
  "value_meaning": {},
  "possible_values": [],
  "data_type": "B"
}
```

Conditions (map)

- condition name
- metadata (if present)
- ID mapping
- …

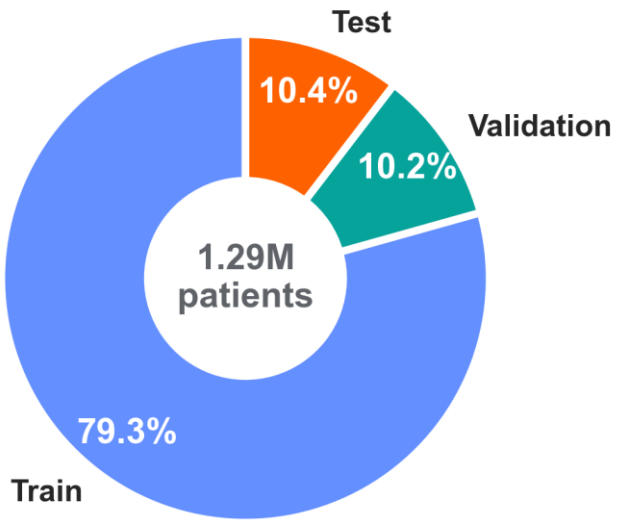
EXAMPLE: PNEUMOTHORAX

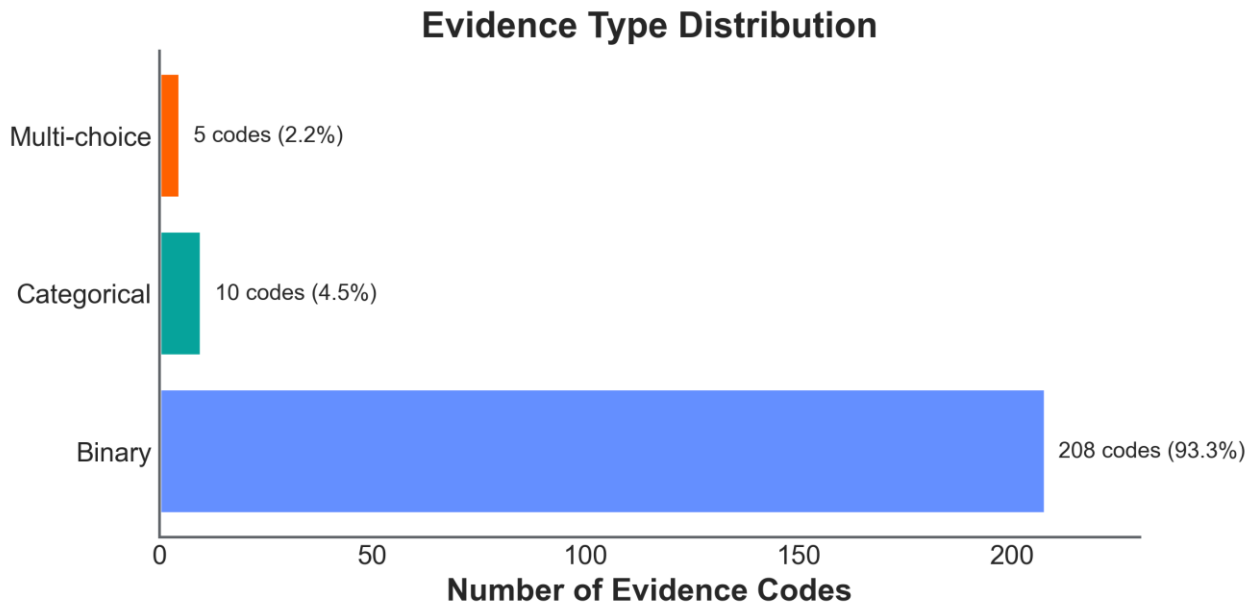
```
{
  "condition_name":
    "Spontaneous pneumothorax",
  "cond-name-fr":
    "Pneumothorax spontané",
  "icd10-id": "J93",
  "symptoms": {
    "E_55": {}, ... },
  "antecedents": {
    "E_79": {},
    "E_165": {}, ...},
  "severity": ...
}
```

Key EDA insight: default values inside EVIDENCES must be filtered, otherwise “symptom presence” is overcounted.

Dataset characteristics

Scale, splits, and evidence schema at a glance





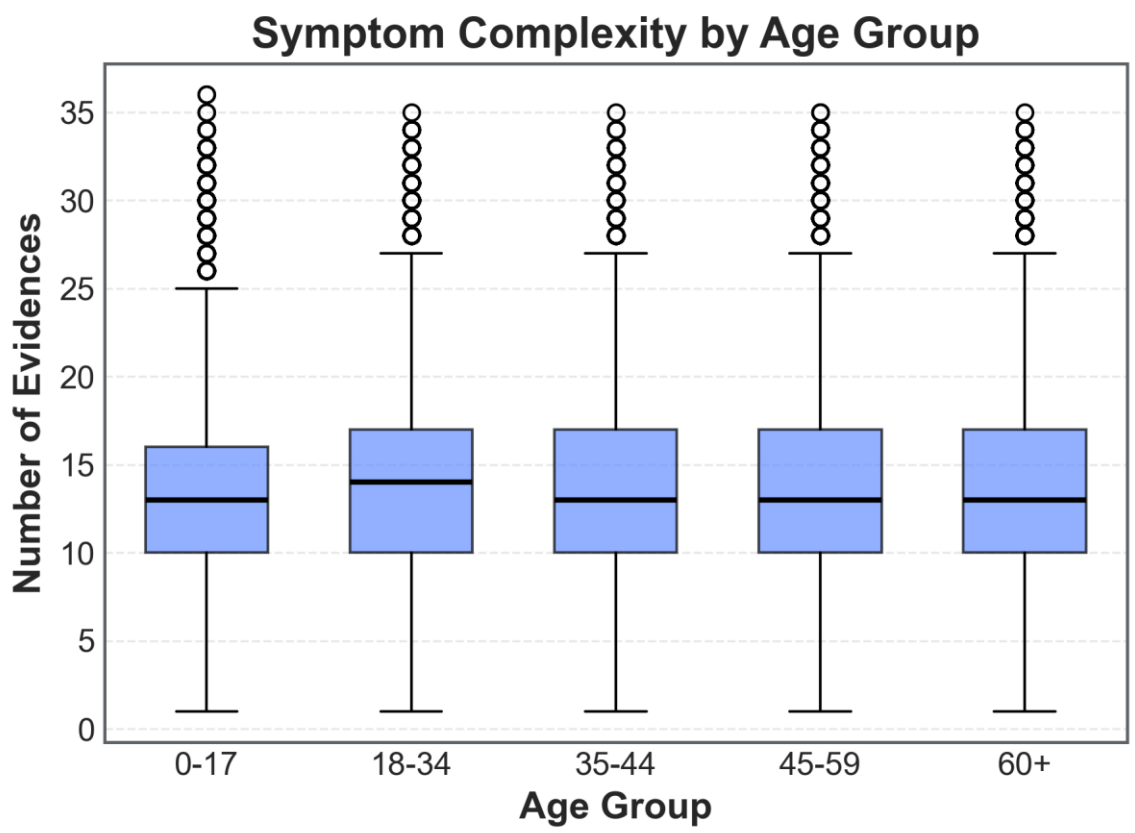
Quick checks:

- 49 diseases (pathologies)
- Sex ~ balanced
- Age 0–109
- Mean DDx length ≈ 9.2

✓ **Scale and quality:** 1.29M diverse cases across 49 conditions with rich evidence types make DDXPlus ideal for supervised learning

Dataset “gotchas”

Three issues that can inflate model performance if ignored



1 **Synthetic limitation**

Age groups show near-identical evidence counts → weaker age-dependent patterns.

2 **Duplicates & leakage risk**

Repeated patterns (~1%) can cross splits and inflate test scores if not controlled.

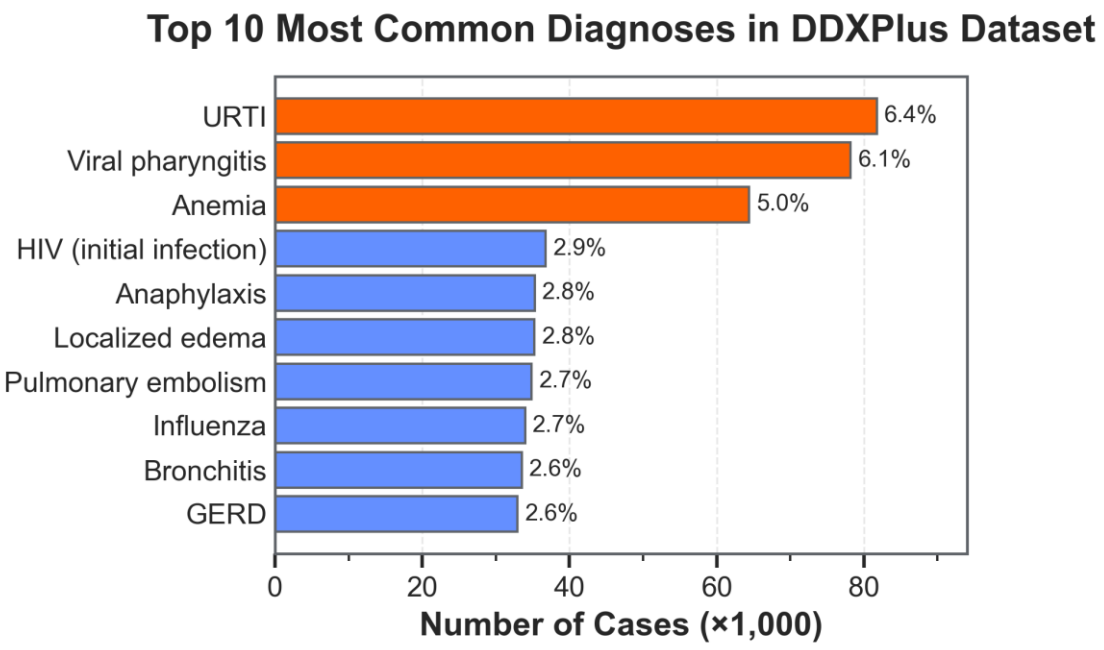
3 **Default values inside EVIDENCES**

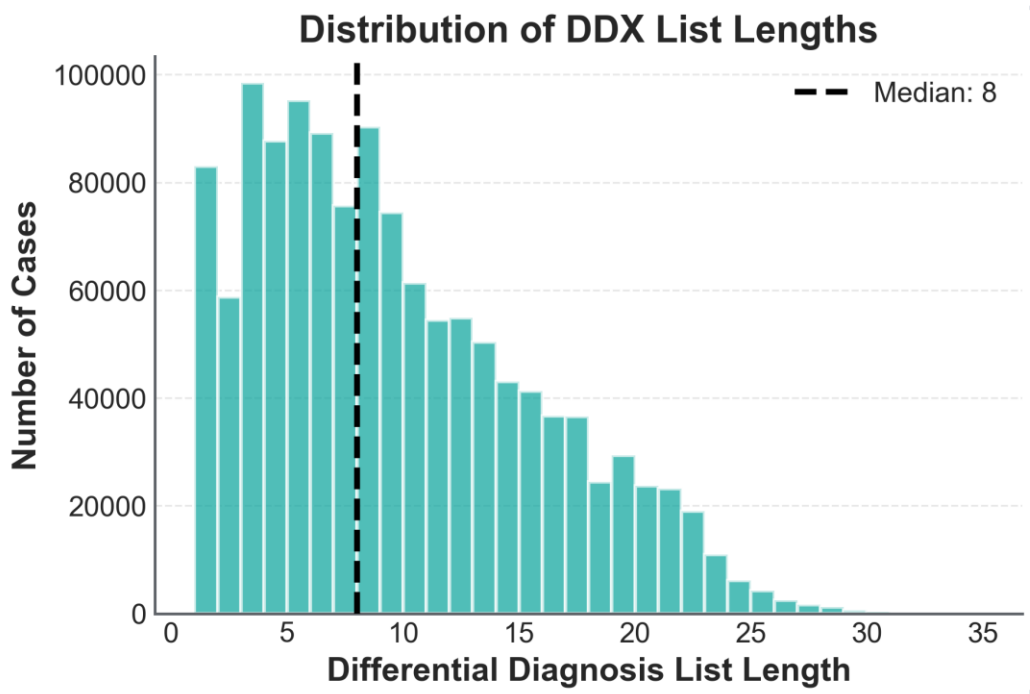
Value-coded evidences include defaults → must filter using per-evidence default_value.

Preprocessing decisions (dedup + leakage removal + default filtering) are essential before training.

Pathologies: imbalance + ranking

Long-tailed labels and variable DDx list length → Top-k evaluation





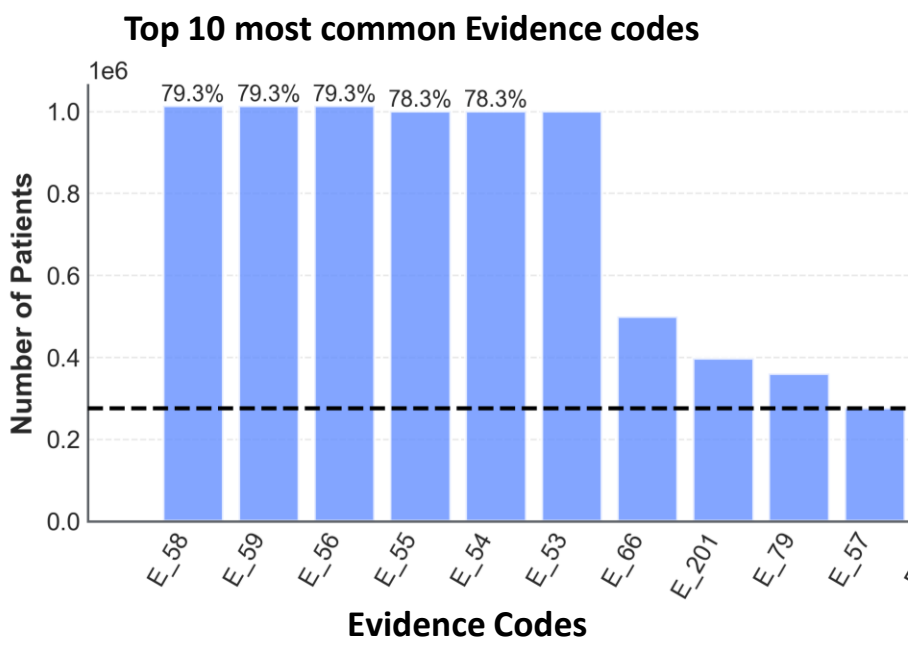
Additional information:

- Disease imbalance: ~254:1 ratio
- Top 3 (URTI, Viral pharyngitis, Anemia) = 18% of all cases
- DDX range: 1-34
- Mean DDx length:9.2, StDev: 6.0

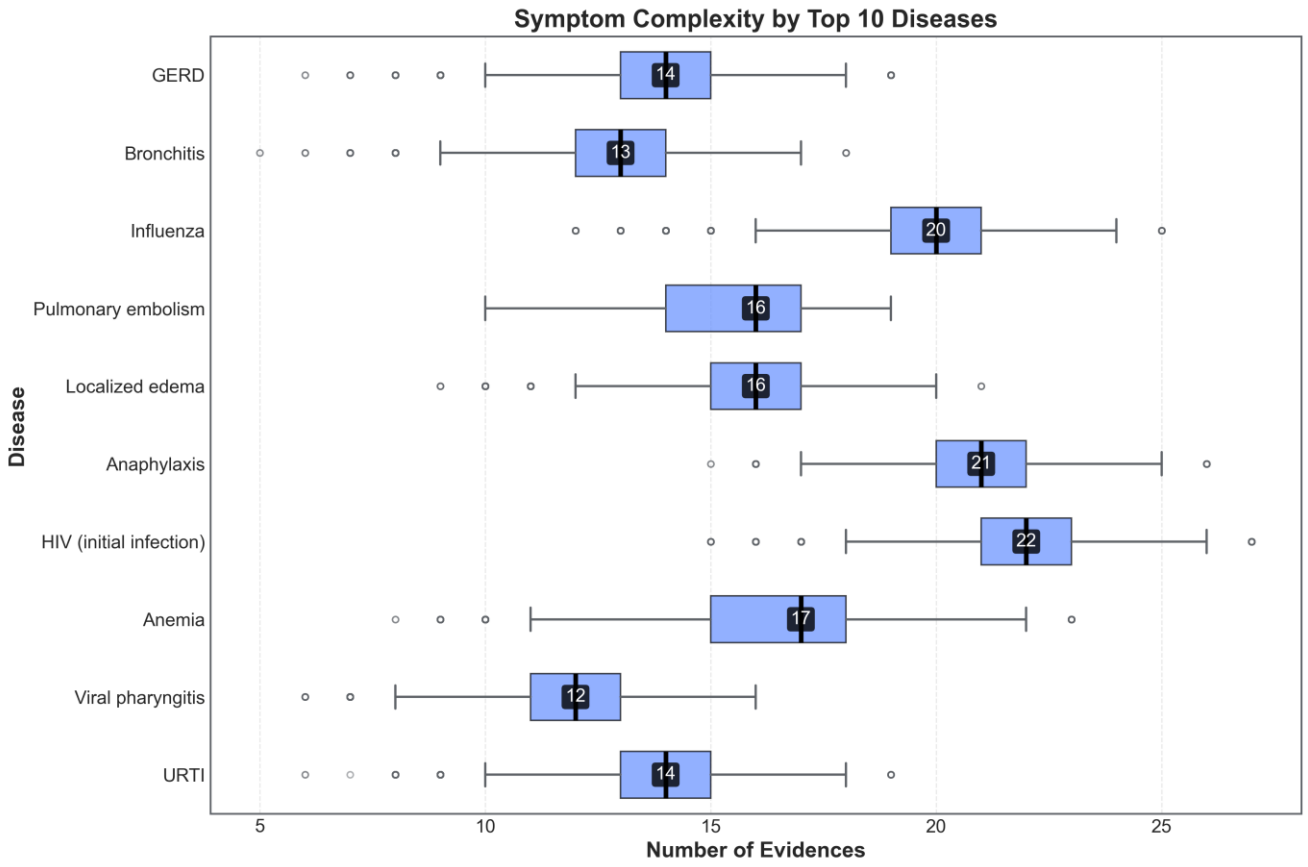
🎯 **Why Top-k matters**: Severe class imbalance (254:1) makes single-label accuracy misleading — ranking metrics (Top-k, NDCG) better reflect real diagnostic workflow where physicians consider multiple plausible diagnoses

Evidence signals

Common questions vs disease-specific patterns



E_58 → "How precisely is the pain located?"
E_59 → "How fast did the pain appear?"
E_56 → "How intense is the pain?"
E_55 → "Do you feel pain somewhere?"
E_54 → "Characterize your pain"
E_53 → "Pain related to consulting?"
E_66 → "Shortness of breath?"
E_201 → "Do you have a cough?"
E_79 → "Do you smoke cigarettes?"
E_57 → "Pain radiate elsewhere?"



💡 **Insight:** Pain-related codes are in 79% of cases (↓ discriminative power), while rare disease-specific codes (E_218, E_175, E_21) show 10-20x ↑signal — guiding our questioning policy to prioritize high-information-gain features

ML Pipeline Overview

From preprocessing to top-k diagnosis prediction

ML Goal: Top-k Diagnosis Prediction

Train a classifier that outputs calibrated probability distributions over the top-k most likely diagnoses given patient symptoms → these predictions feed into the RAG questioning loop.

1. Preprocessing:

- Filter default values
- Feature engineering (binary/categorical)
- Train/test split with stratification

2. Baseline Models:

- Dummy classifiers (Most Frequent, Stratified)
- Bernoulli Naive Bayes
- Logistic Regression (baseline performance)

3. Improved Model (planned):

- XGBoost with hyperparameter tuning (future work)
- LogReg proved sufficient for project scope

4. Base vs Token Comparison:

Compare feature sets:

- Base features only
- Token-level features

5. Calibration & Evaluation

- Probability calibration (Platt scaling)
- Top-k accuracy evaluation
- Confusion matrix analysis

6. RAG Preparation:

- Wrap model in prediction functions
- API-ready interface for RAG loop
- Output: get_top_k_diagnoses

Pipeline outcome: Logistic Regression provides calibrated top-k probabilities → RAG uses these to select high-value follow-up questions → iterative refinement until diagnosis confidence threshold met

Preprocessing translated EDA findings into a safe ML dataset

Leakage control, deduplication, default filtering, and stable feature construction

🔒 Key Preprocessing Steps

🔐 1. Leakage Control

Used `case_hash_inputs` to prevent near-duplicate cases from leaking across train, validation, and test sets

🗂️ 2. Exact Duplicate Removal

Removed exact duplicates with `case_hash_record`; frequency retained only as optional sample weight, never as a feature

🔍 3. Value-Coded Evidence Filtering

Filtered evidences using each evidence's `default_value` from release_evidences.json

✨ 4. Transparent Feature Blocks

Simple, transparent features: multi-hot evidences, one-hot SEX, scaled AGE, and three engineered complexity counts

Feature vector used by the models

Evidence block
base = 223 or token = 972

SEX OHE

AGE

Engineered numerics:
num_items_effective · num_bases_effective · extra_multi_values

💡 **Preprocessing outcome:** Clean, leak-free dataset with transparent features → models can focus on true symptom patterns, not data artifacts → foundation for reliable top-k diagnosis predictions

Why these baseline models?

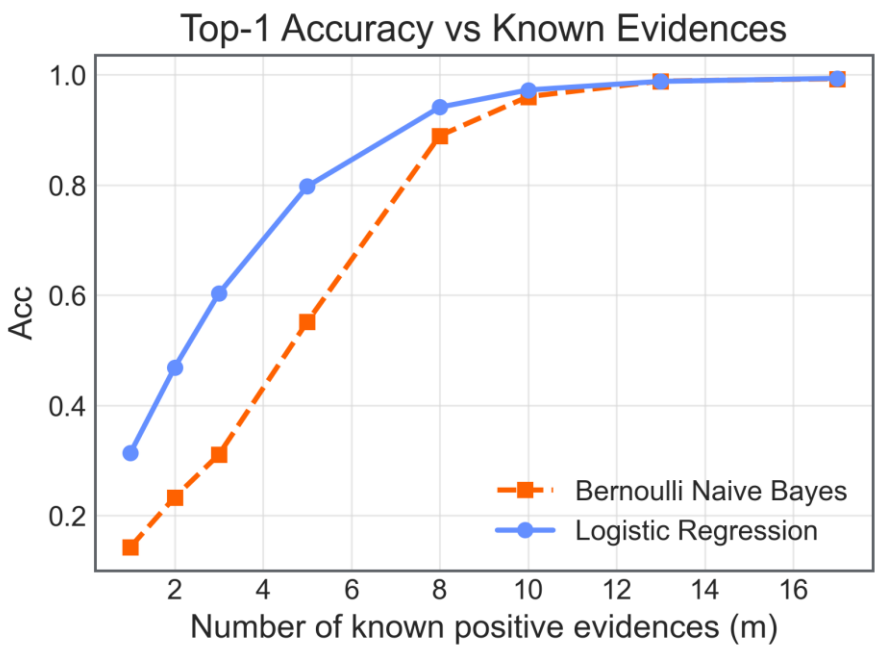
Two sanity checks to set the floor, then two real classifiers suited to sparse binary features

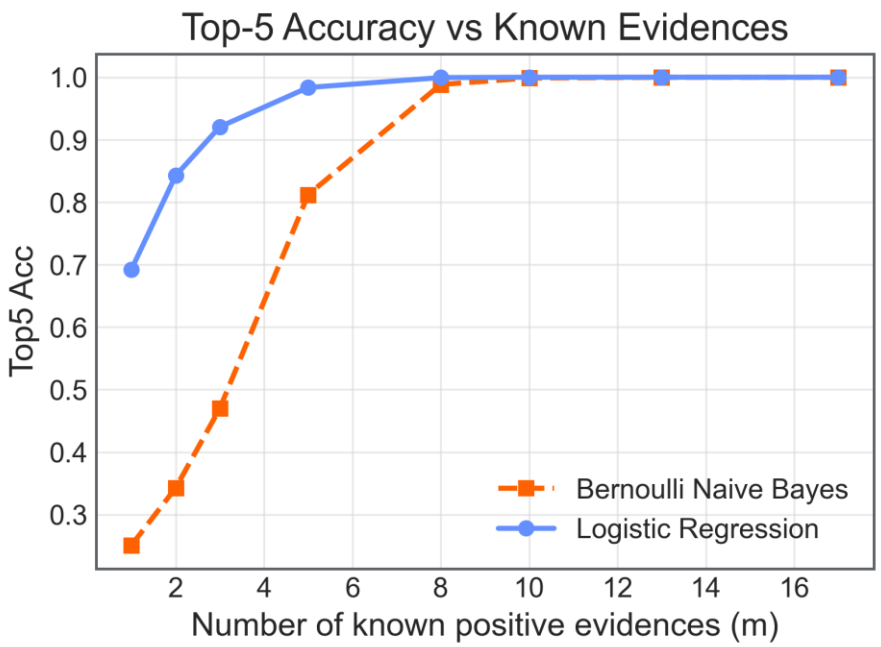
Dummy (uniform)
Sanity floor
for random guessing

Dummy (most frequent)
Sanity check
for class imbalance

BernoulliNB
Classic sparse baseline
fast and simple

Logistic Regression
Strong linear baseline
interpretable coefficients



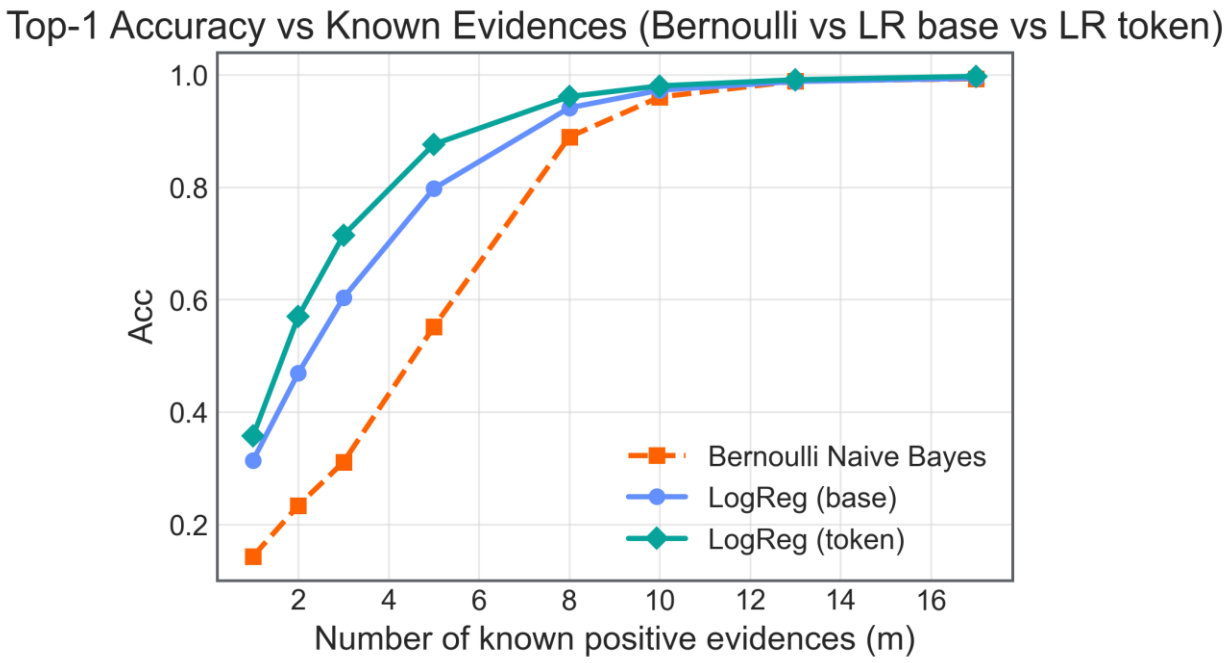


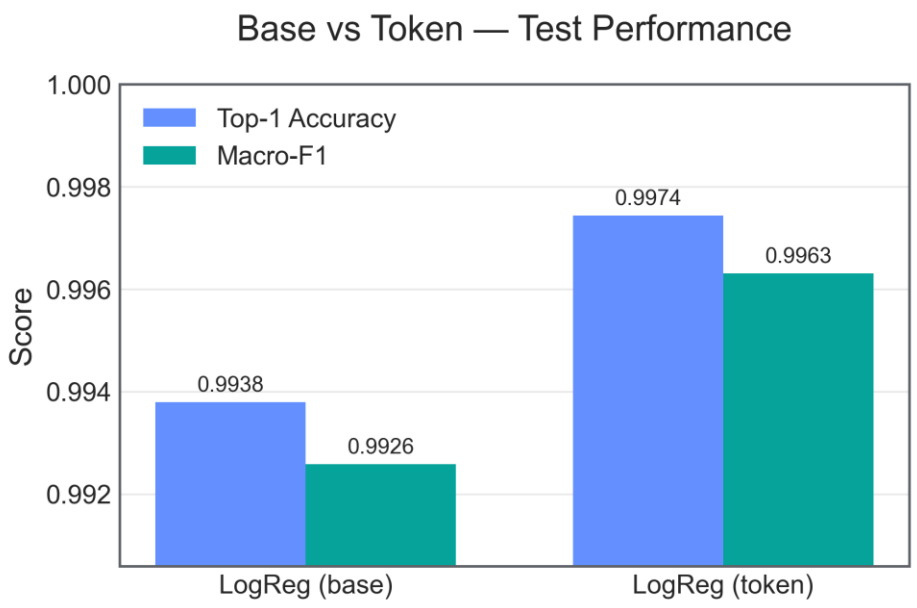
Why LogReg became the primary model

- LogReg outperformed BernoulliNB even with very few known symptoms.
- It remained interpretable through feature coefficients.
- It became the right platform for testing whether a richer encoding — base vs token — could improve the model further.

Token encoding became the main improvement

Value-level evidence preserved detail that base encoding collapses





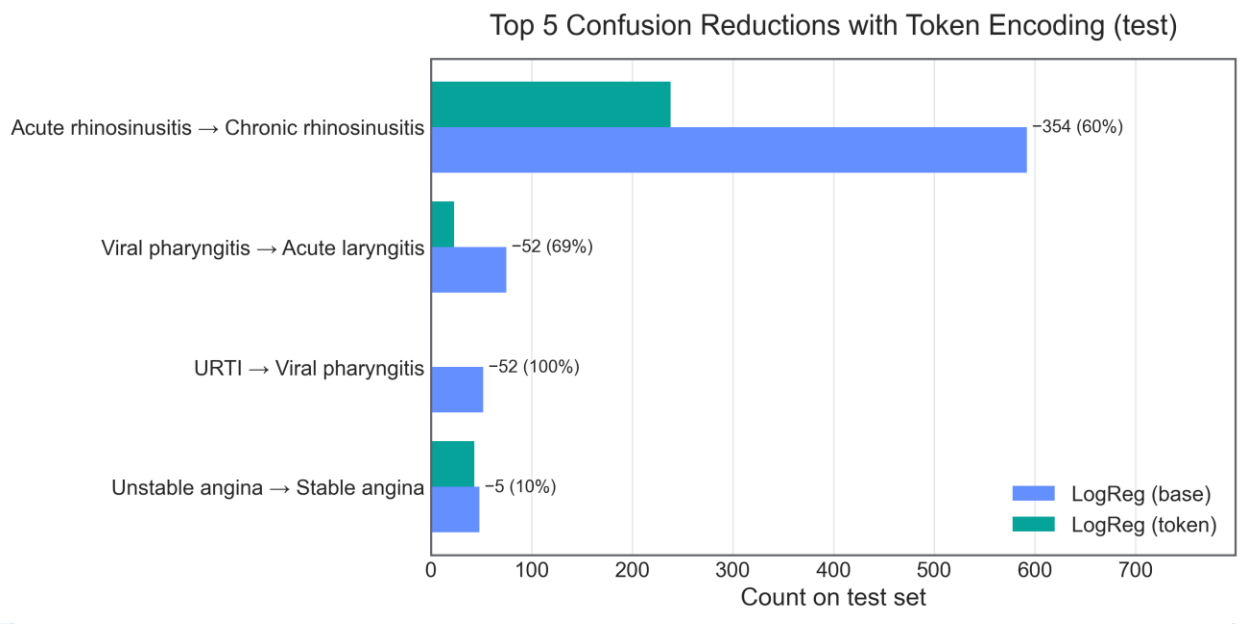
💡 **Key idea:** base encoding keeps only “which evidence exists”, while token encoding also keeps “which value was observed”.

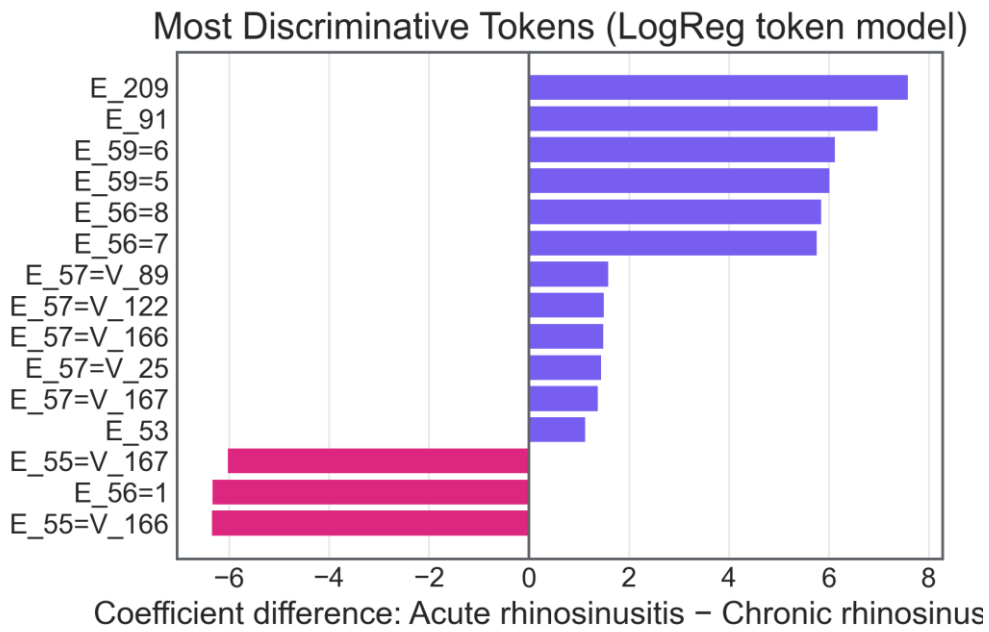
Headline improvement on the full test set

Top-1 accuracy: 0.9938 → 0.9974 · Macro-F1: 0.9926 → 0.9963. The gain is small in absolute terms, but important because performance was already near the ceiling.

Error analysis: where token encoding helped

Confusion reductions and coefficient differences kept the model interpretable

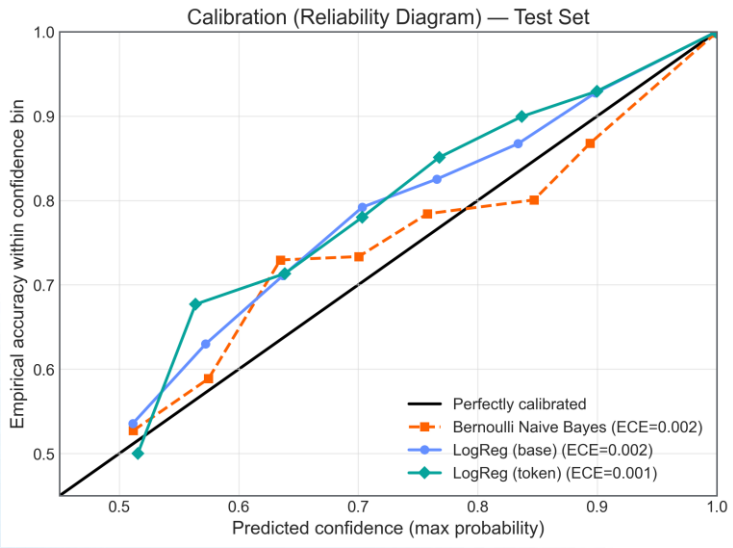


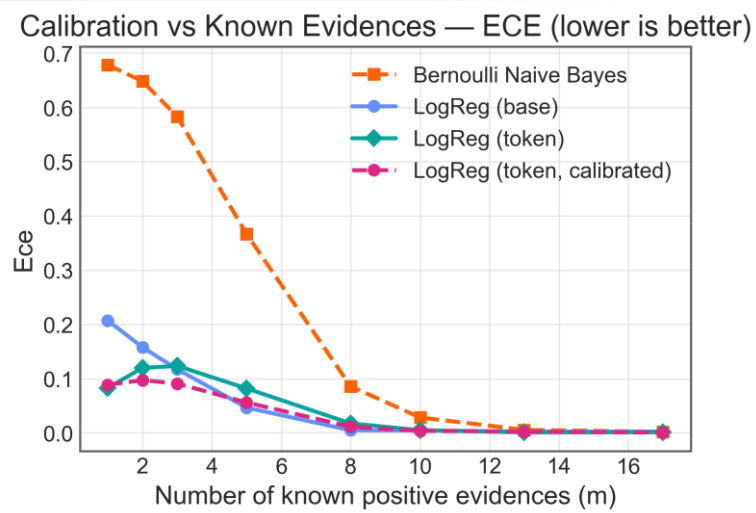


Example: acute rhinosinusitis → chronic rhinosinusitis dropped dropped by 60% (592 → 238 cases). Token coefficients reveal which specific symptom values distinguish this pair.

Calibration prepared the model for policy and RAG

ECE, entropy, and information gain matter because the assistant needs meaningful probabilities





ECE (Expected Calibration Error)

Average gap between predicted confidence and actual correctness. Lower is better; 0 would be perfect.

Entropy

Uncertainty in the predicted disease distribution — high entropy = model is unsure, low entropy = confident prediction.

Information gain

Expected reduction in entropy after asking a question. The next question is chosen to reduce uncertainty as much as possible.

Handoff to the next phase

Saved model artifacts + src wrappers now support the loop: session → encode → calibrated probabilities → next question → RAG explanation.

RAG brings grounded explanations without external APIs

Local TF-IDF retrieval over curated mini-manual + evidence definitions + condition metadata

💡 Why RAG for This Assistant

🎯 Calibrated probabilities need context

Probabilities alone don't explain *why* a diagnosis is likely. RAG retrieves supporting evidence from trusted sources.

🔒 No external API calls = no data leakage

All retrieval happens **locally** using pre-built TF-IDF index. No PubMed queries, no LLM calls during inference.

📚 Explainability without retraining

Mini-manual built offline, retrieved on-demand. Model stays frozen — explanations adapt to query context.

⚡ Fast + deterministic

TF-IDF retrieval is **instant** (<10ms per query). No latency, no rate limits, fully reproducible.

Local Retrieval Corpus

49	223	49
Mini-manual docs	Evidence definitions	Condition metadata

What gets retrieved:

- `explain_topk_diagnoses()` → condition summaries, typical symptoms, severity
- `explain_next_question()` → why this evidence reduces entropy, what it reveals

Saved artifacts:

`outputs/rag/tfidf_retriever.joblib`

`outputs/rag/mini_manual/*.md` (49 docs)

🚫 **Design constraint:** No LLMs in prediction or explanation retrieval. Optional post-processing rewriting layer could be added later (strictly separated from core logic).

✅ **RAG outcome:** Calibrated probabilities + grounded explanations → assistant can justify its reasoning without hallucination risk → educational tool ready for MVP demo

The assistant loop: predict → ask → explain (per turn)

Each iteration refines the differential diagnosis using information gain and retrieves supporting context

1

Update session state

Collect patient age, sex, and answered evidences (positive findings). Negative responses also update posterior probabilities.

2

Predict Top-k diagnoses

Token LogReg (calibrated) outputs disease probabilities. Top-k ranked by likelihood. Probabilities are **meaningful** thanks to calibration.

3

Select next base evidence (information gain policy)

Choose the base evidence that maximally reduces entropy across Top-k. Uses precomputed `policy_artifacts.joblib` — no retraining.

4

Retrieve supporting context (TF-IDF)

Query local corpus: `explain_topk_diagnoses()` for disease summaries, `explain_next_question()` for evidence rationale. Instant retrieval.

5

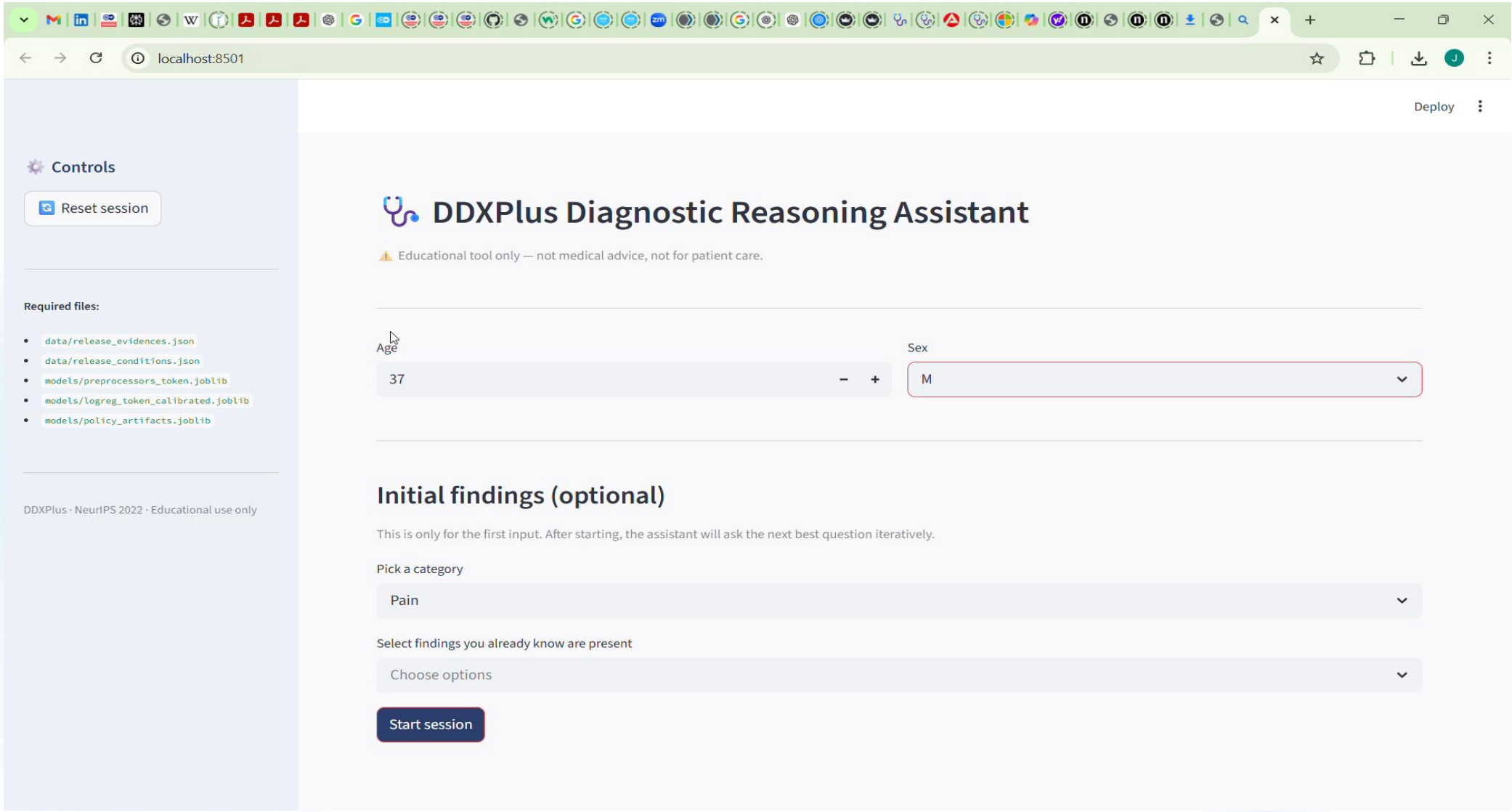
Render explanations (Streamlit UI)

Display Top-k with probabilities + retrieved snippets. Present next question with context. User answers → loop repeats.

🔄 **Loop benefits:** Iterative refinement mirrors clinical reasoning → each question reduces uncertainty → explanations keep user informed → safety-first design with local-only retrieval

Streamlit app deployed

Symptoms' clustering improves user experience



Limitations and what comes next

A strong baseline now; a clear roadmap for improvement after submission

Achieved ✅

- Complete offline pipeline: symptoms → calibrated Top-k → grounded explanations.
- Interactive multi-turn questioning through the Streamlit MVP.
- Grounded explanations using local TF-IDF retrieval and mini-manual enrichments.

Current limitations ⚠️

- DDXPlus is synthetic and limited in scope compared with real clinical settings.
- Model comparison limited (no XGBoost) due to time constraints—feature engineering stayed within MVP scope.
- Evidence phrasing needs refinement; session export feature not yet implemented.

Next steps 🚀

- Refine evidence phrasing and grouping (e.g., pain, respiratory, GI symptom clusters).
- Add session export: downloadable JSON plus a simple educational summary.
- After submission, compare additional models and feature-engineering ideas under the same evaluation setup.
- If an LLM is added later, keep it limited to rewriting and communication rather than prediction.

Closing takeaway: The MVP demonstrates explainable, multi-turn diagnostic reasoning—future work focuses on evaluation rigor and UX refinement.

"One more thing..."

Try the assistant yourself.

- Enter symptoms → get Top-k differential diagnoses
- Calibrated probabilities update in real time
- Grounded RAG explanations per diagnosis
- Answer one question → watch the list refine

💬 Questions? Let's discuss!

🐙 **github.com/JuliaP-create/diagnostic-reasoning-assistant**

📱 SCAN TO EXPLORE THE CODE

Full source code · README · Notebooks
Models · Streamlit app

`github.com/JuliaP-create/diagnostic-reasoning-assistant`

Educational tool only · Not medical advice